

CSE331 - Automata and Computability

Quiz 1

Section 16 (Set A)

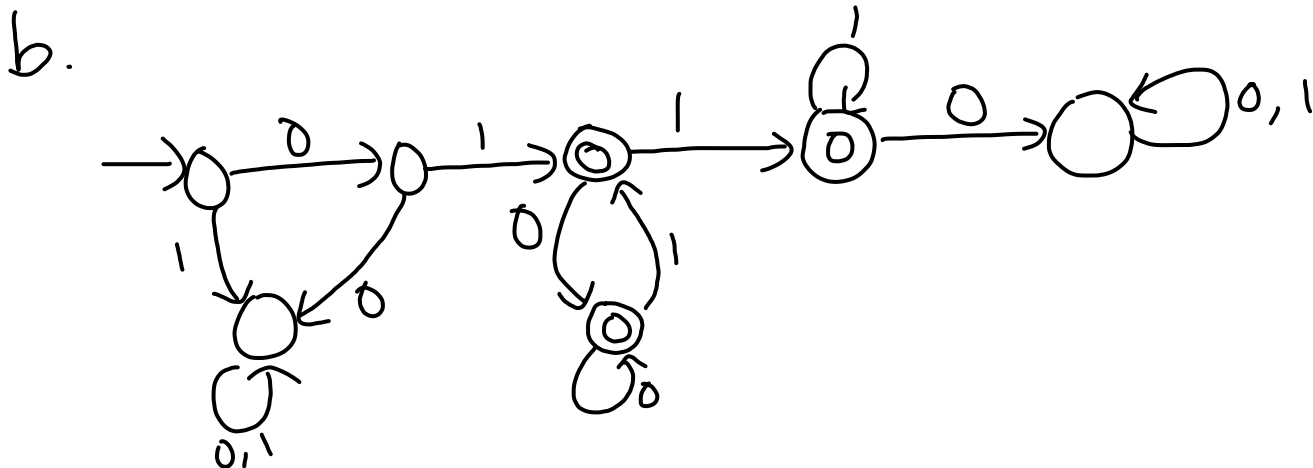
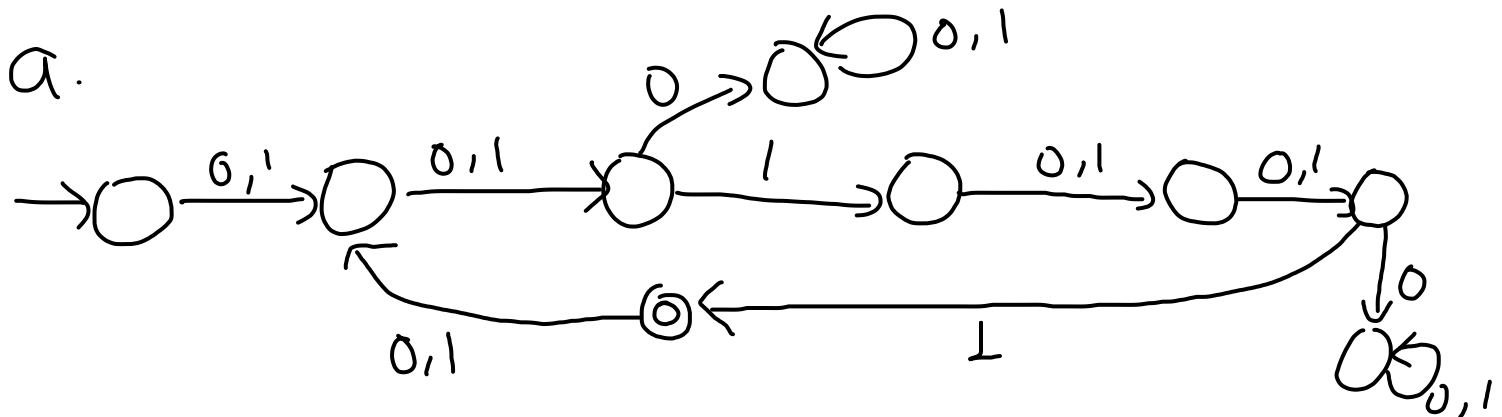
Full name:

ID:

Total Marks: 20

1. (5*2 marks) Draw the DFA of the following, $\Sigma = \{0, 1\}$:

- Non-empty string with length as multiple of 6, every $6k$ and $6k+3$ position will contain 1.
- Starts with 01 and doesn't contain 110.



2. (3+3+4 marks) Answer the questions based on the following, $\Sigma = \{0, 1\}$:

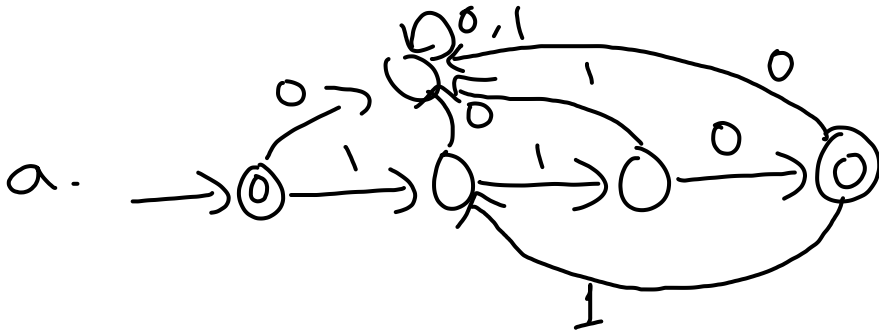
$$L_1 = \{110\}$$

L_2 = The first and third position will contain the same symbol if they exist.

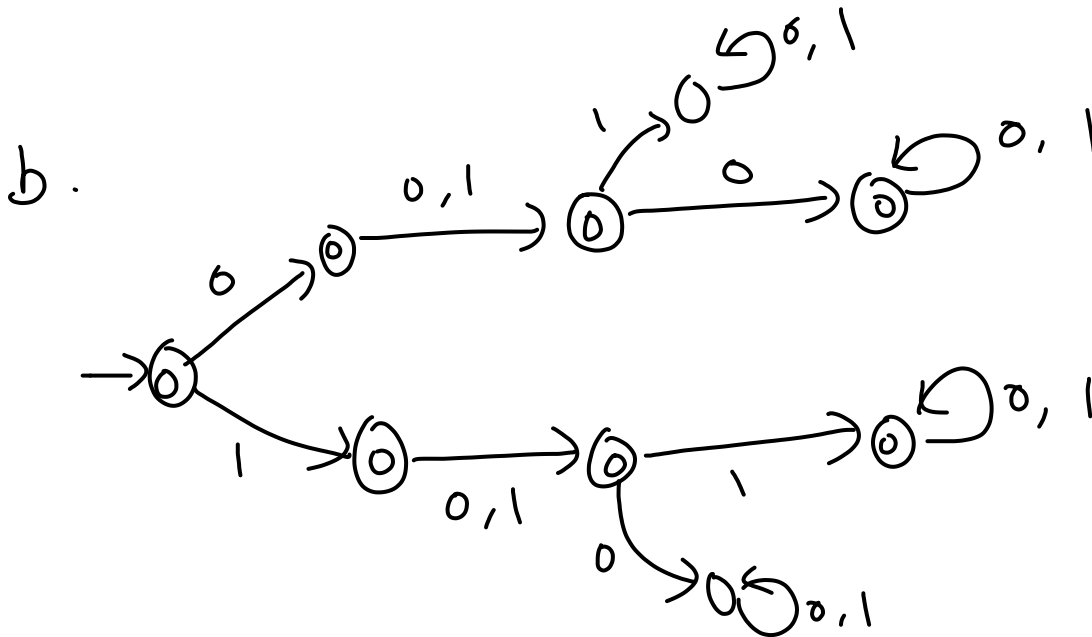
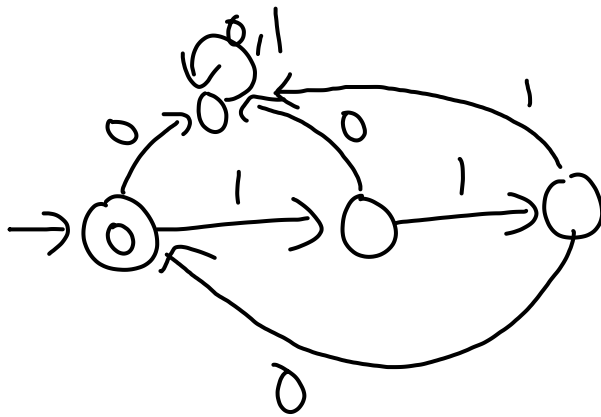
a. Draw DFA of L_1^*

b. Draw DFA of L_2

c. Draw the complete transitions of DFA of at least 5 states of $L_1^* \cup L_2$ using the cross-product method



OR,



CSE331 - Automata and Computability

Quiz 1

Section 16 (Set B)

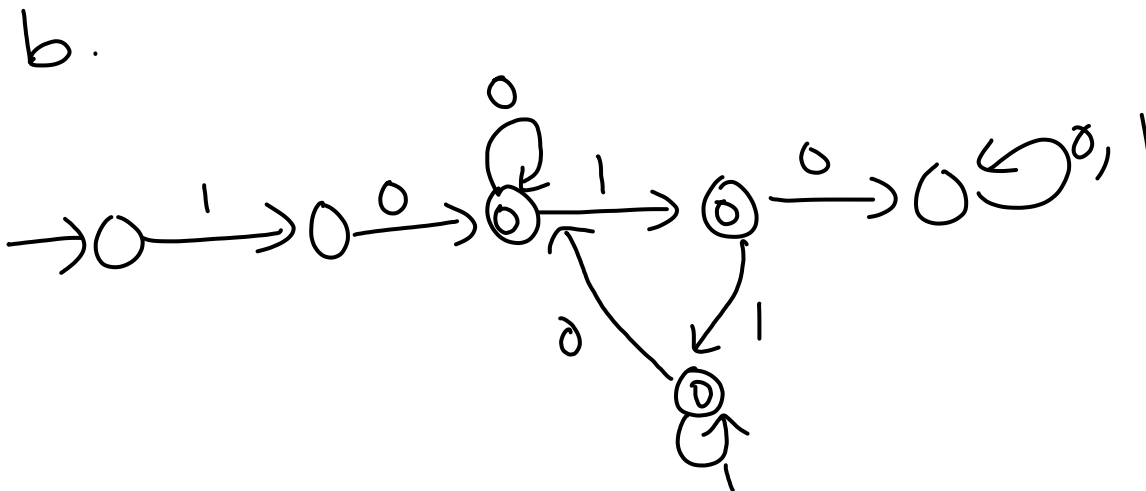
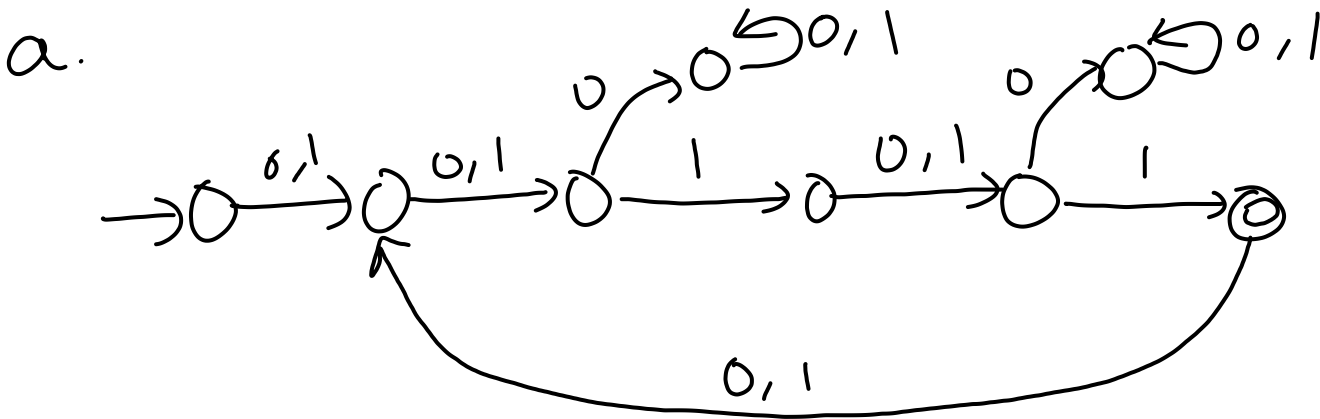
Full name:

ID:

Total Marks: 20

1. (5*2 marks) Draw the DFA of the following, $\Sigma = \{0, 1\}$:

- Non-empty string with length as multiple of 5, every $5k$ and $5k+3$ position will contain 1.
- Starts with 10 and doesn't contain 010.

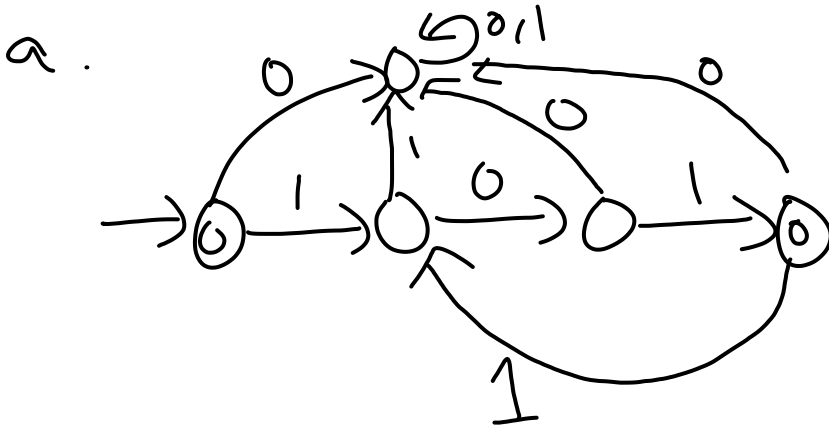


2. (3+3+4 marks) Answer the questions based on the following, $\Sigma = \{0, 1\}$:

$$L_1 = \{101\}$$

L_2 = The first and third position will contain different symbols if they exist

- Draw DFA of L_1^*
- Draw DFA of L_2
- Draw the complete transitions of DFA of at least 5 states of $L_1^* \cup L_2$ using the cross-product method



OR;

